Name: Shristi Keshri

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Instructor: Naomi Zweben

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Robot-assisted surgery, often referred to as robotic surgery is a method where doctors perform critical and complex surgeries with the assistance of robots. It provides a higher level of precision, flexibility, and command over the surgery which would be difficult to achieve with the conventional methods [1].

Robotic surgery has been rapidly growing due to its beneficial medical implications. However, the initial purchase cost and continual maintenance are major economic issues that cannot be ignored as they have stinted the growth of the operation worldwide [5]. In 2021, the market of robotic surgery in the United States (US) was 35% of the worldwide market. However, it is disappointing that over six billion people all over the world have very limited access to the advantages of robotic surgeries [3]. For instance, robotic surgeries in India are in a nascent state with an overall contribution of less than 0.1% on the global scale [6]. Taking these factors into account the best ethical system to evaluate the conundrum is Act Utilitarianism to measure whether the benefits are worth the costs and possible repercussions.

Robotic surgeries were considered a futuristic idea that could hardly become a reality [2]. However, the application of robotics in surgery can be dated back to the

1980s when the technology called ROBODOC was used to perform hip replacement procedures and PROBOT found its application in urological surgeries [4]. Then in the 1990s, thorough research was conducted by the US National Aeronautics and Space Administration (NASA) and Stanford Research Institute where they investigated the potential for robotics in telepresence surgery or remote surgery which would enable surgeons to perform remotely in different geographical locations [1, 4].

As a result of all this research and advancements in technology, today, robots assist in more than 14 million procedures all around the world [3]. In addition to that, more than 60,000 surgeons worldwide have been trained in the da Vinci systems (Intuitive Surgical, Sunnyvale, CA, USA) as it is the only acceptable robotic surgical unit that is used to perform minimally invasive surgeries. This robot is a sophisticated system that uses precise movements and improved vision to perform minute incisions with minimal blood loss [2, 9]. Looking at the projections from 2012 to 2018, the application of robots in surgical procedures has increased from 1.8% to 15.1% [2]. These percentile increases are observed as the use of technology in surgery has extensive medical benefits. In comparison to conventional surgeries, robot-assisted surgeries make sure that patients have less pain during recovery [7]. With the precise and complex tools used in the surgery, there is minimal blood loss during the procedures, lowering the risk of infections. In addition to that, it guarantees fast-paced recovery and thus ensures shorter stays in the hospital [7].

In contrast to all the medical benefits that robotic surgery provides, the option is not always economically viable. To prepare for a robotic surgery, the respective hospital is required to have the da Vinci Surgical System [9]. However, the system has been a matter of discourse due to its high cost of operations since it was approved by the United States Food and Drug Administration (US FDA) in 2001. Each of these units' purchasing cost ranges from $1 million to $2.5 million, in addition to the maintenance and consumable prices which could be estimated to be around $3,500 to $5,000 for each surgery [3, 8].

Furthermore, even though robotic surgery is designed in a way that would minimize human errors, it is not foolproof. Individuals are to undergo rigorous training to ensure that these robotic surgical training programs can significantly decrease surgical errors [11]. According to the Cleveland Clinic, surgeons are required to finish the basic training and additional specialized training to perform robot-assisted surgeries which could last for 1-3 years [7]. Thus, in addition to medical school, which according to the AAMC, amounts to an average of $268,476 for resident students and $363,836 for students who attend private schools [10], these specialized surgeons are required to take robotic surgery training which costs about $500/day according to The College of Medicine of The Ohio State University [9].

Despite countering the human limitations by training and availability of robotic surgery systems in hospitals, another prevailing disadvantage is the maintenance costs of the equipment. If not kept in check, robotic malfunctions, though rare could occur which may result in an alteration to the planned surgical procedure. Mechanical failures could also risk the life of the patient being operated costing the hospital a lot more than money [11].

From what has been discussed hospitals are asked to weigh the medical benefits with their financial security. Ethically, it would be difficult to consider a rule that would solve every hospital's dilemma and there are too many extraneous factors (malfunctions, manual errors, etc.) to simply consider measuring net positivity. Thus, a middle ground should be considered with Act Utilitarianism. According to Act Utilitarianism proclaimed by Jeremy Bentham and George Stuart Mill, an action is good if its benefits exceed its harms and bad if the harms exceed its benefits. In the medical field, ethics plays a very crucial role. It requires dealing with ethical and legal dilemmas of the patients on their decision directives about their lives like their decisions about wealth [12]. In the context of robot-assisted surgeries, hospitals nationwide have invested billions of dollars in the required machinery and spent a great deal to correctly train the surgeons who perform these procedures. These investments are a lot. However, the medical benefits are long-lasting. As stated above, robotic surgeries have advantages ranging from rapid recovery time to precisely performing complex surgeries with minimal blood loss [11].

On the other hand, according to the Strategic Market Research report, the market for robot-assisted surgical systems in North America reached USD 2.8 billion in 2021, and by 2028, the market will reach USD 7 billion, growing at a 14.9% CAGR [3]. As observed, the United States of America is the leading country in robotic surgery, and it makes up 35% of the worldwide market and 85% of the market in the region [3]. In contrast to that, the robotic surgeries performed in India constitute only 0.1% of the total surgeries performed worldwide [11].

The cost disparities between conventional and robotic surgeries are more pronounced in India in comparison to the USA. These nations have completely different healthcare budgets. On one hand, the USA spends 17% of its GDP on facilitating a proper healthcare system – the highest per capita expenditure on healthcare worldwide, while India just spends 4-5% of its GDP [13]. Out of this budget, a small portion is used for robotic surgeries making it very rare in India. Another disadvantage India faces is the implementation of the National Health Policy (NHP). Though it is a very ambitious program initiated by Prime Minister Narendra Modi, its success depends on the ruling political parties of the different states as the medical advancements may be hampered by political interests [13]. Additionally, most of the resident training in India is in the nascent stages and requires fellowship programs as the training provides minimal exposure to robotic surgeries. Further, the higher costs involved with the surgery make it a far-fetched possibility for developing countries like India [11].

To help resolve this issue, in 2021, Dr Srivastava designed India's first surgical robot named SSI Mantra. The main idea for this system was that it would be a cost-effective method to help the general masses achieve high-quality medical care as it could be performed at one-third the cost of current da Vinci systems. However, according to Dr. Pradeep Jain, Principal Director and HOD, Gastrointestinal Oncology, Fortis Hospital, India is lagging behind the United States and the SSI has very limited usage. "Some instruments for such surgeries cost Rs 3 to 4 lakh ($3,600 to $4,800), but their usage is limited to just 10 procedures, and the cost also ranges from Rs 1 to Rs 1.5 lakh ($1,000 to $1800)," he said [6]. Addressing these prevailing monetary, political, accessibility, and awareness issues in India requires coordinated efforts from healthcare policymakers and what their motives are for the technological development in surgeries for their country.

Considering the summation of the surgical benefits and monetary harms, according to the principles of Act Utilitarianism nations should be encouraged to let this technology evolve a few more years and then expand the market so that more robot-assisted surgeries can be performed at a high success rate for lower costs.

In conclusion, the adoption of robotic surgeries considers the medical benefits, economic repercussions, and ethical implications. While robot-assisted surgical procedures have positive patient outcomes, precise work on complex cases, and flexibility, they also entail a large amount of investment in equipment, training, operations, and maintenance. There is also a prevailing disparity between nations like the USA and India where robotics is at a different level of advancement. Applying the principles of Act Utilitarianism, we saw that the long-term and fast-paced medical advantages outweigh the economic investments, and thus the awareness and accessibility of robotic surgeries should be increased. Although this will require a sustained commitment and patient approach from industrial stakeholders, with the successful implementation of resources the full potential of robotic surgeries can be harnessed which would benefit our society.

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    This article on The Future of Robotic Surgery was published by the Engineering Research Department of Vanderbilt University. It gives a detailed description of what robot-assisted surgery is and what the future of that field is ranging from micro-robotics to AI to telepresence. The source had an understandable definition which helped me get my paper started by building its foundation. As this article was published on November 9th, 2023, it is recent and hence relevant. In addition to that, it is published by one of the reputed schools in the United States of America and the information is reliable.

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3. Strategic Market Research. (2023). *Top robotic surgery statistics to follow in 2023*. Top Robotic Surgery Statistics to Follow in 2023. https://www.strategicmarketresearch.com/blogs/robotic-surgery-statistics

    This article is from the Strategic Market Research website posted in August of 2023. This resource was remarkably helpful for my research

paper as it provided a lot of statistics from what was the market like for robotic surgery in the USA, India, the UK, and several other countries to how many da Vinci systems are in the countries. It assisted in the analysis of how the different countries, in my case the United States and India are progressing in the field. The source follows an intensive research methodology to obtain precise and accountable results. They conduct primary and secondary research and publish only validated information by the higher-ups. Thus, I chose to use this source due to its reliability and relevance.

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   This paper is published by Oxford Publishing and written by Elizabeth Z. Goh and Tariq Ali. Elizabeth is a faculty of Medicine at the University of Queensland, Brisbane, Queensland, Australia, and Tariq is from the Department of Oral and Maxillofacial Surgery, Royal and Brisbane and Women’s Hospital, Brisbane, Queensland, Australia. These qualifications make them and this editorial paper a reliable source. This paper shows how robotic surgery is progressing and takes numerous disciplines into consideration like general surgery, urology, head surgery, and more. I used this resource to support my claim of how beneficial robot-assisted surgery is and the wide advantages of the practices prevailing in the field. This paper is also from 2022 making it relevant today for usage.

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This article published by The Week Magazine is about the lack of awareness of robotic surgeries in India. This magazine has been around for years, and its articles are an informative and illuminating read. It was also written by the respectable journalist Akanki Sharma on April 2nd, 2023. The piece talks about how India is in its nascent stages of robot-assisted surgeries and gives accurate percentages for the arguments. It was also helpful for my paper as I incorporated the direct quotations by Dr. Pradeep Jain and the ideas of SSI by Dr. Srivastava, both mentioned in this article. I was also able to use the cost estimations for the surgeries in India as it helped me compare them with those of the USA in the global section of my research paper.

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This article was published by the National Library of Medicine in August 2023 by authors Satvik N Pai, Madhan Jayaraman, Naveen Jayaraman, Arulkumar Nallakumarasamy, and Sankalp Yadav. All these authors are reputable surgeons working in the orthopedic and general medicine fields in the Hospital for Orthopedics, Sports Medicine, Arthritis, and Trauma (HOSMAT) Hospital and Shri Madan Lal Khurana

Chest Clinic in India. In this article, we see how robotic surgery is a fast-growing field that has evident contributions to the surgical landscape. It talks about the robotic training required by the surgeons and how that can be improved which was very helpful in my paper. I also used the information about how robotic malfunctions even though rare, if occur could be a major disadvantage of the field. It also mentions the high cost involved with these procedures making it very difficult for developing countries like India to access these resources.

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